

# KALPANA T

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## PROFESSIONAL SUMMARY

Dedicated Electronics and Communication Engineering student with a keen interest in embedded systems, IoT, and hardware design. Demonstrates strong analytical, communication, and teamwork skills, along with a proactive approach to learning new technologies. Eager to contribute technical expertise and gain practical experience through challenging projects and opportunities.

## EXPERIENCE

### Embedded Systems Internship at Manfree Technologies

Dec 2025

*Onsite*

- Involved in Product Requirement Analysis, Embedded Programming, Hardware Design etc.,
- Worked on real-time interfacing of sensors and actuators.

## PROJECTS

### Digital Alarm clock using RTC and LCD| *Arduino IDE, Embedded System*

Sep 2024

- Designed and developed an RTC-based digital alarm clock using Arduino, DS3231 RTC module, LCD display, and buzzer.
- Integrated user interaction through push-button controls for setting and managing alarm timing efficiently.

### Classroom Automation System | *IoT - ESP32, Embedded C*

Dec 2025

- Developed a Classroom Automation System using ESP32 and Embedded C.
- Implemented automated control of lights and fans using IoT technology.
- Integrated sensor interfacing and Wi-Fi communication for smart monitoring and control.

### UWB Antenna for Breast Cancer Detection| *GNU Octave, Simulation*

May 2026

- Performed simulation and analysis using GNU Octave for UWB signal propagation and detection.
- Applied concepts of antenna design, microwave imaging, and biomedical signal analysis in the

## TECHNICAL SKILLS

- **IoT:** Sensor Integration, Cloud Monitoring, Network Communication Protocols
- **Embedded Systems:** Microcontroller Programming, Processor-Based Systems, Wireless Communication
- **Electronics Design:** PCB Layout Design, Circuit Design and Simulation
- **Design Software:** Canva
- **Engineering Tools:** Keil µVision, Arduino IDE, STM32CubeIDE, LTspice, Proteus, KiCad, GNU Octave
- **Programming Languages:** Python, C, Embedded C, Java

## CORE COMPETENCIES

- Communication Skills, Time Management, Critical Thinking, Problem Solving, Teamwork and Collaboration, Adaptability

## EDUCATION

### Coimbatore Institute of Engineering and Technology

2023 - 2027

BE. Electronics and Communication Engineering

8.75 CGPA

### Scism Matric Higher Sec. School

Mar 2023

HSC

78.8 %

## CERTIFICATIONS

Embedded System Design using C      *Maven Silicon*

Verilog HDL-Hands On      *Maven Silicon*

VLSI SoC Design using Verilog      *Maven Silicon*

Semiconductor- VLSI & Embedded Systems      *Maven Silicon*

## DECLARATION

I hereby declare that the information provided above is true and correct to the best of my knowledge and belief.